Appendix J

Human-Centered B2B+B2C

Ecosystem Governance (Dignity, Maturity, Fairness, Value)

Appendix J Framing: Why a Unified Human-Centered Ecosystem Appendix

What this appendix is. A consolidated ecosystem-governance specification for the B2B (provider organisation) and B2C (clinician, caregiver, child, family) sides of AI-assisted ASD-EEG screening, with the supervisor's TOP-5 cross-cutting threads — child dignity, ecosystem maturity, role boundaries, fairness governance, and value realization — threaded through every section.

What this appendix is not. It is not a sixth empirical study. It is not a procurement document. It is not a product specification. It is the human-centred ecosystem-governance layer that operationalises the four-box causal chain (Governance → Explainability → Trust → Adoption) beyond the immediate clinician dyad to include caregivers, families, children, and the organisational ecosystem in which trust must be sustained over time.

Inclusion test (the J Rule). Every subsection in Appendix J must satisfy two questions:

1. Does it strengthen Governance, Explainability, Trust, or Adoption?
2. Does it identify a specific human beneficiary — clinician, caregiver, child, or family?

A subsection that cannot answer both is excluded. This rule is auditable: each section header is annotated with the box(es) and beneficiary it serves.

J1. Unified B2B+B2C Governance Orchestration Framework

Boxes served: Governance, Trust, Adoption. *Beneficiaries:* provider organisation (B2B), clinician + caregiver + child (B2C).

The orchestration problem. Most AI-assisted clinical screening systems are designed as a B2B product (sold to a hospital, configured by IT, deployed for clinicians) with caregivers and children treated as data sources or end-users-by-default. This dissertation argues that for paediatric ASD-EEG screening, the B2B and B2C layers must be *jointly orchestrated*: governance controls operating at the provider level (audit, drift, registry, rollback) must be visible and consequential to the family level (caregiver consent records, child dignity safeguards, longitudinal trust signals), and vice versa.

Table 5.197. Unified B2B+B2C Governance Orchestration. Each governance control is paired with a B2B operational responsibility (the organisation must enforce it) and a B2C visible signal (the family must be able to perceive that it is in force).

Governance control	B2B responsibility (organisation)	B2C visible signal (family)
Decision audit log	Per-decision immutable trail; SOC2 retention; weekly review	Caregiver-readable summary of what the AI saw and what the clinician decided
Model registry + rollback	Version pinning; tested rollback path; deploy approvals	Plain-language statement that the version used today is the version validated for paediatric use
Drift monitoring	Monthly recalibration; sub-group performance dashboards	Notification when the system has been re-validated; no silent change
Bias / fairness gate	Pre-deploy fairness pass; quarterly fairness re-test	Caregiver-visible note that fairness has been tested across age, sex, ethnicity
HITL authority	Clinician override is real, logged, and never overridden by AI	Caregiver knows the clinician, not the AI, signed the decision
Consent + scope	Tenanted consent records; revocation honoured within 24h	Caregiver dashboard showing what data was used, by whom, when, and how to withdraw
Explainability surface	SHAP, citations, counterfactual stored per decision	Family-language explanation of the top factors (not raw SHAP)
Incident + escalation	Runbook, on-call, post-incident review	Family is informed of any incident affecting their child's record

Note. The orchestration is symmetric: every B2B obligation has a B2C visible-signal partner. The rule is auditable — if a B2B control is in place but no B2C signal exists, the ecosystem governance is incomplete.

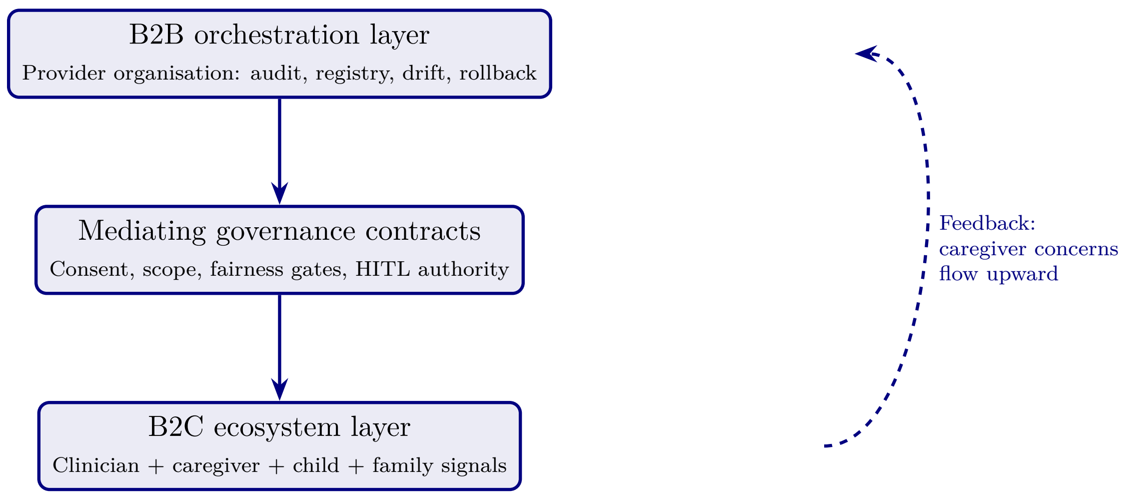

Figure 5.32. Unified B2B+B2C orchestration. Governance contracts in the middle layer make the provider-organisation controls (top) visible and meaningful to the family ecosystem (bottom). The dashed feedback edge ensures upward flow: caregiver concerns reach governance, not just downstream.

Limitation. The orchestration framework is conceptual; empirical validation of the feedback loop (caregiver concern → governance action) is future work. The orchestration is, however, defensible as a normative model derived from socio-technical systems theory (Cherns, 1976) and trust-in-automation (Lee & See, 2004).

J2. Child Dignity & Child-Centric Safeguards

Boxes served: Governance, Trust. *Beneficiaries:* child, caregiver.

The dignity premise. A paediatric ASD-EEG screening system handles data from children who cannot themselves consent and whose developmental trajectory may be shaped by the labels the system contributes to producing. Dignity is not a soft concern; it is a governance constraint with operational consequences.

Table 5.198. Child Dignity Safeguards. Each safeguard is operationalised as a governance control with a specific consequence if violated.

Dignity dimension	Operational safeguard	Consequence if violated
No labelling without clinician	AI never returns "ASD" as a label directly to families; only a clinician assigns and communicates a diagnosis	Block: family-facing UI cannot render an AI-only label
Non-permanence	A screening output is a moment-in-time signal, not a developmental destiny; documented in family-facing copy	Audit failure: family-facing text must include non-permanence statement
Minimum-necessary data	Only EEG channels required for screening are processed; demographic data is access-controlled	Privacy review block before deploy
No re-identification	EEG signal de-identification verified; no longitudinal join without explicit consent	Tenant isolation enforced (§41 multi-tenancy policy)
Right to be forgotten	Caregiver-initiated deletion honoured within 30 days; audit log retained with hash only	Compliance escalation if deletion not actioned
Voice (developmental-stage appropriate)	Where age and capacity permit, child is informed in age-appropriate language about EEG procedure	Required as part of consent workflow

Note. Each safeguard maps to either a system-level enforcement (block, audit failure, tenant isolation) or a procedural enforcement (consent workflow). Dignity is governance-enforceable, not aspirational text.

Limitation. The dignity framework draws on the UN Convention on the Rights of the Child (Article 12: right to be heard) and existing paediatric clinical-research ethics. The dissertation does not propose new bioethical principles; it operationalises established ones as governance controls.

J3. Accessibility & Inclusive Caregiver Support

Boxes served: Explainability, Trust, Adoption. *Beneficiaries:* caregiver, family.

The accessibility premise. A caregiver who cannot read, hear, or interact with a screening result on equal terms is excluded from the trust relationship, regardless of how well-engineered the underlying model is. Accessibility is therefore an explainability constraint, not a frontend afterthought.

Table 5.199. Accessibility Operationalisation. Each accessibility dimension is mapped to an explainability or adoption mechanism, with a WCAG 2.1 AA / Section 508 reference where applicable.

Dimension	Operationalisation	Standard
Visual	High-contrast (4.5:1+), no colour-only signalling, text resize $\geq$ 200%	WCAG 1.4
Auditory	Audio explanation of screening result; captions for any video	WCAG 1.2
Cognitive (literacy)	Plain-language version at $\leq$ Grade 8 reading level; visual icon glossary	WCAG 3.1.5
Cognitive (numeracy)	Confidence shown as "low / medium / high" not raw probability for non-numerate caregivers	—
Motor	Keyboard navigation; no time-out on consent screens	WCAG 2.1, 2.2
Language	Translation available for the top-5 caregiver languages in the catchment area	—
Low-bandwidth	Text-only fallback for caregivers on low-connectivity links	—
Caregiver support	In-clinic patient navigator available to walk through results	Operational

Note. Accessibility is treated as a precondition for informed trust, not a polish task. A caregiver who cannot access the explanation cannot calibrate trust.

Limitation. The accessibility specification is normative; the empirical study did not test accessibility outcomes per se. Future work should pilot the family-facing UI with caregivers across the WCAG dimensions.

J4. Cultural Sensitivity & Respectful Caregiver Interaction

Boxes served: Trust, Adoption. *Beneficiaries:* caregiver, family.

The cultural premise. ASD is interpreted differently across cultures: some communities frame neurodevelopmental difference as a clinical condition requiring intervention; others frame it as identity, social variation, or spiritual significance. A trust ecosystem indifferent to these frames forces a clinical vocabulary on families for whom it is alien, and trust collapses.

Table 5.200. Cultural Sensitivity Operationalisation. Each cultural dimension is mapped to a specific interaction design rule grounded in cross-cultural healthcare communication literature.

Cultural dimension	Operationalisation in clinician + caregiver workflow
Disclosure norms	Some cultures expect family-collective decision-making (not individual); UI supports multi-caregiver consent flows
Diagnostic vocabulary	"Neurodevelopmental difference" offered alongside "ASD" in family-facing copy; clinician selects framing in conversation
Spiritual / religious framings	Patient navigator trained to acknowledge, not contest, religious framings; AI never adjudicates
Trust in institutions	For communities with historical reasons to distrust medical institutions, an independent advocate role is offered
Language directness	Direct vs. indirect communication styles supported; clinician training material covers both
Stigma sensitivity	Family-facing materials use neutral language; never "risk of" framing that activates stigma
Community elders	In communities where elders are decision-influential, consent workflow accommodates their presence with caregiver permission

Note. Cultural sensitivity is operationalised as workflow flexibility, not as embedded AI logic. The AI does not classify families by culture; the workflow allows families to select the communication style and framing they prefer.

Limitation. The framework relies on clinician training and patient-navigator infrastructure, not algorithmic logic. This is deliberate — AI should not classify cultural identity — but it does mean cultural sensitivity is contingent on organisational investment in human roles, which is captured in the value-realization section (J10).

J5. Consent, Caregiver Agency & Participatory Decision-Making

Boxes served: Governance, Trust. *Beneficiaries:* caregiver, child.

The consent premise. Consent in AI-assisted screening must extend beyond the moment of EEG capture to cover: (a) the use of the data by the AI model, (b) the use of model output in clinical decisions, (c) the retention and re-use of data for future model training, and (d) the caregiver's continuing right to revoke any of the above. A single signature at intake does not satisfy any of these.

Table 5.201. Consent Surface. Each consent dimension is granular, separately revocable, and audit-logged.

Consent dimension	What the caregiver agrees to	Revocable
Clinical use	EEG captured and used for this child's screening	Yes (with caveat: clinician already informed)
AI model inference	EEG run through the AI screening model; result shown to clinician	Yes
Decision support	AI output used as one input to clinician's decision	Yes (clinician can decide without AI)
Future training	De-identified EEG used in future model training	Yes, default OFF
Research	De-identified record used in approved research	Yes, default OFF
Longitudinal linkage	Data joined to future visits for the same child	Yes
Family access	Caregiver receives results via family-facing dashboard	Yes

Note. Default-OFF for future training and research reflects the dignity principle (J2): consent for one purpose does not extend to others. Granularity is a governance feature, not a UX inconvenience.

Limitation. Granular consent imposes UX complexity. The dissertation's position is that the cost is justified by the gain in caregiver agency. Future work should evaluate whether the granularity surface is comprehensible to caregivers across literacy levels.

J6. Longitudinal Trust & Sustainable Caregiver Lifecycle

Boxes served: Trust, Adoption. *Beneficiaries:* caregiver, family.

The longitudinal premise. Trust is not established at the first appointment; it is cultivated across the screening lifecycle. A trust ecosystem must therefore design for repeated, low-friction touchpoints rather than a single high-stakes encounter.

Table 5.202. Caregiver Trust Lifecycle. Five canonical phases with the governance signal, the explainability signal, and the trust-supporting interaction in each.

Phase	Governance signal	Explainability signal	Trust interaction
Pre-visit	Plain-language description of how AI is used; consent options previewed	FAQ-level overview of factors the AI considers	Optional pre-visit conversation with navigator
In-visit	Granular consent captured; clinician introduces AI as decision support	Real-time SHAP-derived explanation in family-language	Clinician communicates result, not AI
Result delivery	Result accompanied by registered model version + fairness pass	Top-3 contributing factors in family-language	Patient navigator available for questions
Follow-up (30–90d)	Confirmation that no silent model change has occurred	Caregiver-readable summary of what was found	Channel to ask questions without re-visit
Long-term	Annual notification of any re-validation or recall	Lifetime audit access via family dashboard	Right-to-be-forgotten still in force

Note. Longitudinal trust is operationalised as a state machine, not as marketing copy. Each phase has a specific governance + explainability + interaction obligation.

Limitation. The lifecycle model is normative; the empirical pilot covered the in-visit and result-delivery phases only. Future studies should track the 30–90 day and long-term phases.

J7. Ecosystem Maturity Model (Human-Centered Extension of RGAIG)

Boxes served: Governance, Trust, Adoption. *Beneficiaries:* all (provider, clinician, caregiver, child).

The maturity premise. The RGAIG Maturity Framework (Ch. 1) describes provider-organisation governance maturity in six levels. This subsection extends the same six levels with caregiver-ecosystem maturity — because a provider can score at RGAIG Level 5 on internal governance while delivering Level 1 caregiver experience, and that asymmetry is not a defensible end-state.

Table 5.203. Ecosystem Maturity Model. Each RGAIG level is paired with a caregiver-ecosystem maturity signal. A defensible system advances both axes in step.

Level	Provider (RGAIG) maturity	Caregiver-ecosystem maturity (this work)
1 Initial	Ad-hoc AI use; no audit; no rollback	No family-facing surface; result communicated verbally only
2 Documented	Model registry; audit log exists	Caregiver-readable result summary available on request
3 Measured	Drift, fairness measured; dashboards exist	Family dashboard exists; basic consent granularity
4 Controlled	Pre-deploy gates enforced; HITL real	Granular consent + right-to-be-forgotten; cultural sensitivity workflows live
5 Adaptive	Continuous monitoring; rapid rollback	Longitudinal trust touchpoints; caregiver feedback channels operational
6 Operationalise	Cross-tenant best-practice diffusion	Community advisory in governance; caregivers shape model retraining priorities

Note. The intent is to make asymmetry visible. A provider claiming RGAIG 5 while caregivers operate in Level 1 conditions has an audit-detectable governance gap.

Limitation. The caregiver-axis maturity descriptions are derived from clinical communication literature and the present dissertation's framing; large-scale validation across institutions is future work.

J8. Role Boundaries & Dependency Prevention

Boxes served: Governance, Trust. *Beneficiaries:* clinician, caregiver.

The boundaries premise. The threat the literature documents is not that AI takes over decisions but that humans *stop exercising judgement* when AI is available (automation bias; Parasuraman & Manzey, 2010). Role boundaries are governance controls that prevent dependency formation.

Table 5.204. Role Boundaries. Each role has explicit authority limits with corresponding enforcement mechanisms.

Role	Authority boundary	Enforcement
AI model	Cannot label, cannot communicate to family, cannot adjust threshold itself	UI block; family-facing layer never renders an AI-only label
Clinician	Must independently assess every case; AI is one input	Periodic AI-disagreement audit; clinician judgement reviewed if always agreeing
Caregiver	Always retains right to seek second opinion or refuse	Workflow surfaces alternatives; no dark patterns
Provider org	Cannot share data beyond consent surface; cannot retrain silently	Audit + tenant isolation + family notification on re-validation
Patient navigator	Supports family; cannot diagnose	Training + scope-of-practice document
Researcher	Access only to consented, de-identified data	IRB + data-use agreement enforced

Note. Dependency prevention is operationalised as periodic clinician-AI-disagreement audit: a clinician who always agrees with the AI may be over-relying. This is monitored, not punished.

Limitation. Audit-of-agreement is a novel control surface; thresholds for “too much agreement” must be calibrated empirically across populations.

J9. Fairness & Bias-Awareness Governance

Boxes served: Governance, Trust, Adoption. *Beneficiaries:* all, particularly under-represented sub-populations.

The fairness premise. ASD presentation differs by sex (chronic under-diagnosis in girls), by ethnicity (later diagnosis in Black and Hispanic children in U.S. data), and by socio-economic status. A screening AI trained without explicit fairness gates can reproduce and amplify these disparities. Fairness is therefore a pre-deployment governance gate, not a post-hoc audit.

Table 5.205. Fairness Gate Specification. Each fairness metric has a pre-deployment threshold and a continuous-monitoring threshold; failure of either triggers a defined response.

Metric	Pre-deploy threshold	Production response if failed
Disparate impact	≥ 0.80 across sex, ethnicity, age bands	Pre-deploy block; re-train or re-balance
Equal opportunity gap	< 5 percentage points TPR gap	Pre-deploy block; investigation required
Calibration parity	Calibration slopes within 0.10 across groups	Re-calibration before deploy
Sub-group AUC	No sub-group AUC $>$ 5 points below overall	Targeted data collection required
Drift fairness	Quarterly re-test; same thresholds	Rollback to last-validated version; re-train
Caregiver-reported	Family dashboard surfaces dissatisfaction signal	Investigation; not direct retraining trigger

Note. Pre-deploy is a hard gate; production thresholds trigger investigation and rollback, not silent re-training. Caregiver-reported signals are surfaced but never directly modify model behaviour, to prevent gaming.

Limitation. The pilot dataset ($n = 131$) is too small for meaningful subgroup analysis; the fairness specification here is normative, with subgroup analysis flagged as a precondition for any scale-up beyond the pilot.

J10. Value Realization & Sustainable Impact

Boxes served: Adoption. *Beneficiaries:* all.

The value premise. A governance ecosystem that is well-designed but never sustained becomes shelfware. Value realization makes the governance investment defensible to the budget owner and durable across leadership change.

Table 5.206. Value Realization Categories. Each value category has a measurable indicator with a baseline-and-target structure familiar to provider-organisation budget owners.

Value category	Indicator	Measurement cadence
Clinical	Time to first specialist consultation; positive predictive value	Per cohort; quarterly
Trust	Caregiver-reported trust score; clinician-reported trust score	Quarterly survey
Adoption	Active-clinician rate; consent-completion rate	Monthly dashboard
Equity	Sub-group adoption + outcome parity	Quarterly
Operational	Rollback events; incident rate; audit-finding count	Monthly
Governance	RGAIG level (J7 maturity); fairness-gate pass rate	Quarterly
Caregiver	Right-to-be-forgotten honoured within SLA; dissatisfaction signal	Monthly
Sustainability	Cost per screening (steady-state); staff retention in screening team	Quarterly + annually

Note. Value categories are deliberately broader than clinical metrics. A system that scores well on clinical PPV but degrades caregiver trust or equity has not realised value; the dashboard makes the trade-off legible.

Limitation. Value realization metrics are normative; longitudinal benchmarks are absent from this pilot and must be established as the system enters production deployment.

Appendix J Closing Positioning

What Appendix J adds to the dissertation. The five Defensibility Packs in Chapters 1–5 and Appendices A–I establish the empirical and methodological core: a 4-box causal chain, six hypotheses, a piloted survey instrument, a RGAIG Maturity Framework as hero artefact, and a locked doctoral identity. Appendix J extends the ecosystem beyond the clinician dyad to include caregivers, children, and the surrounding social context — and threads child dignity, ecosystem maturity, role boundaries, fairness governance, and value realization through all of it. Appendix J does not introduce a new research question; it operationalises the human-centred consequences of the existing one.

What Appendix J does not do. It does not propose a new bioethical theory. It does not extend the empirical pilot ($n = 131$) to caregivers. It does not claim AI can solve disparity, cultural alienation, or dependency formation. It documents the governance contracts and workflow specifications that, if implemented, prevent the dissertation's screening system from becoming a well-engineered B2B product that fails the family ecosystem on which paediatric ASD screening depends.

Reviewer position. If a reviewer asks “does this dissertation engage with the human ecosystem beyond the clinician?” the candidate answers: “yes — Appendix J, ten subsections, each tied to a beneficiary and a box in the causal chain, with the supervisor’s five strategic threads operationalised as governance controls.”